

4-Week Plan: Momentum V1 & God Mode

Week 1: The Core Engine (Momentum V1 Foundations)

- Day 1-2: Pandas & Vectorization Basics. Learn how to load stock data without writing slow for-loops. Understand how column-based math works in Pandas.
- Day 3-4: The 6-Month Momentum Factor. Learn why 126-day returns work. Master the `df['Close'].shift(126)` function to calculate past performance while strictly avoiding look-ahead bias.
- Day 5-6: Data Masking & Cross-Sectional Ranking. Learn how to filter out penny stocks (`df['Volume'] > min_vol`) and use `.rank(pct=True)` to instantly sort 500 stocks from best to worst.
- Day 7: Equal Weighting. Understand the logic behind distributing your capital evenly (5% each) across the Top 20 stocks.

Week 2: Defense & Regime Filtering (Bridge to V2)

- Day 1-3: Simple Moving Averages (SMA). Learn how to calculate rolling averages (`.rolling(window=200).mean()`).
- Day 4-5: Building the Regime Filter. Write the logic that looks at the Nifty 50 Index. If the Nifty drops below its SMA 200 (for V1) or SMA 100 (for God Mode), trigger the Go to Cash signal.
- Day 6-7: Periodic Resampling. Learn how to use Python to isolate the last trading day of the month so you know exactly when to run your monthly rebalance.

Week 3: Advanced God Mode Math (V2 Specifics)

- Day 1-2: Jegadeesh-Titman (12M - 1M) Momentum. Understand the academic research behind why skipping the most recent month of returns prevents mean-reversion losses. (Formula: Close 21 days ago / Close 252 days ago).
- Day 3-4: Realized Volatility. Learn how to calculate the standard deviation of daily returns (`np.std`) and multiply it by `sqrt(252)` to get annualized volatility.
- Day 5-6: Inverse Volatility Weighting. Instead of equal weights, learn how to allocate more capital to safe, slow-moving stocks, and less capital to wild, volatile stocks.
- Day 7: Leverage Mechanics. Understand the math behind applying 1.1x leverage to a heavily concentrated Top 5 portfolio.

Week 4: Statistical Hardening (Robustness Testing)

- Day 1-2: Out-of-Sample & Walk-Forward Testing. Learn why testing on data your model hasn't seen yet is critical. Understand how to slice Dataframes by specific years.
- Day 3-4: Monte Carlo Gaussian Noise. Learn how to use `numpy.random.normal()` to inject random fake slippage and bad execution prices into your backtest to see if the strategy survives.
- Day 5-6: Pseudo-Random Trade Shuffling. Learn how to use `numpy.random.permutation()` to scramble the order of all

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your historical trades. If the strategy still makes money when the trade order is shuffled, you have a true mathematical edge.

- Day 7: T+1 Execution Modeling. Code the ultimate reality check: generating the signal at the close of Day 1, but actually buying at the open price of Day 2.